Math **163** Name (PLEASE PRINT) Srinarayan Srikanth

Exam **1A** **2026** Fall/Spring/ Summer I/**Summer II** Score 0

Please fill in the following table putting in expected points corresponding to the parts of problems that you have completed and submitted and an expected total. Leave blank those parts that you have not completed or that do not exist. You will lose double the points that a part is worth if you submit incomplete, unproved, or unchecked answers, or if you do not follow the directions of the problem. Ideally you should know exactly what you have done correctly and how many points you should receive for parts and in total. I will fill in the received grade after I correct the exam. Not all questions have the same number of parts so that some columns may be redundant for particular problems.

										Expected	Received
	Part	Part	Part	Part	Part	Part	Part	Part	Part	Grade	Grade
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)		
Number											
1a											
1b											
1c											
2a											
2b											
2c											
3a											
3b											
3c											
								Total			

Instructions

- All work must be done using a **MECHANICAL PENCIL following instructions given**.
- If you have any questions at all, please **ask me**. This includes the following possibilities:
    - something is **illegible,** or **ambiguous**, or **unclear** to you.
- **You may use your homework, notes, books, and calculators unless otherwise stated during the exam.**
- **You will not get any partial credit**. You have to show **absolutely all your work** on all questions, **exactly as in the notes, with no changes, variations, or omissions in form, unless otherwise noted whether orally or in writing**. If your answer is correct, but your procedure is incorrect or is not allowed by the instructions, you will get no credit all. All work must be displayed using **correct syntax**, and all **explanations**, written in **complete sentences**, must be complete.
- **You must write neatly, clearly and legibly. You must draw every straight-line segment other than those in the list: (**$=, \neq, \equiv, <, \leq, >, \geq, +, -, \times, \div, \ \neg, \vee, \wedge, \parallel, \perp, \backslash, \rightarrow, \leftrightarrow, \mapsto$**) with your ruler. At least 1 point will be taken off for each instance of violation. Signs of equality or equivalence must line up, one below the other on the left of expressions. Every symbol has a specific meaning and may be used only in this sense and in no other. Syntax defined in the notes must be followed exactly to the letter.**
- Remember that 'does not make sense' is a possible answer.
- I can grade you only on what you **actually** put on paper; I cannot read minds. You cannot say later that what you had written down is not what you had meant. **You need to learn to write exactly what you mean and mean exactly what you write**.
- **The answers for a problem should be noted on the page on which the problem appears in the areas indicated. All work for the problem should be shown on sheets that follow the page on which the problem appears numbered as follows. Work for a problem that occurs on page n should be on blank (not ruled) sheets after page n, numbered, n0, n1, n2 etc. on the lower right-hand corner as in this document. For example, work for a problem on page 502 should be on pages numbered 5020, 5021, 5022 etc. as needed. Work that corresponds to different parts must be separated from each other by striaght lines drawn with a ruler parallel to the edges of the paper and labelled correctly as in the Pages from Student Work as found under Files on CANVAS.**
- **You should submit a SINGLE document.**
- **You may NOT include pages that you have NOT attempted**.
- **As shown in the class means as shown in the notes.**

D Define $Y := Yes$ $N := No$

R **Note that in a set an element may occur exactly once**. Therefore, there may not be any repeated elements in any set in the following pages. **So, on every problem, if you do not have the correct sets, all your answers will be completely wrong and you will not het any points at all. Even though in the notes I have had to use parentheses to circcle Y and N in the computation of sets, please circle them.**

R If answers to different instances of some question on the exam contradict each other you will get a score of zero for every instance.

R **You will lose points for making any extraneous mark on the exam.**

R You may only submit complete answers to questions marked explicitly on the exam. **You will lose points for unfinished work or incomplete attempts. In particular all proofs and refutations must follow exactly the format that is found in the notes and must contain every detail.**

R Separate pieces of work for any part of any problem recorded on a blank page intended for the purpose must be indexed by the index of the problem and also by numbers starting at 0 that record the order in which the pieces were completed; such parts must be separated from each other by straight lines parallel to the edges of the paper drawn with a ruler. You willx lose points for otherwise.

R Any sloppiness, untidiness, and any departure from proper format (as indicated in class) will lead to a score of 0.

R **Please do not expect any points if you do not follow the instructions to the letter.**

D $\forall x \in \mathbb{N}$ $Prm(x) :\Leftrightarrow x\ is\ prime$

D $\forall x \in \mathbb{N}$ $Odd(x) :\Leftrightarrow x\ is\ odd$

D $\forall x \in \mathbb{N}$ $Evn(x) :\Leftrightarrow x\ is\ even$

6 (1a) Answer the following questions exactly as shown in class:

Your J-number $JN :=$ _____ $:= (J_k | k \in 1..8)$

2 (i) $A := \left\{ J_k \in 0..9 \middle| J_k \geq k \right\} = \left\{ \quad \right\}$

k		**1**		**2**		**3**		**4**		**5**		**6**		**7**		**8**	
J_k																	
$J_k \geq k$		***Y***		***Y***		***Y***		***Y***		***Y***		***Y***		***Y***		***Y***	
		N		***N***		***N***		***N***		***N***		***N***		***N***		***N***	

2 (ii) $B := \left\{ J_k \in 0..9 \middle| k < J_k \right\} = \left\{ \quad \right\}$

k		**1**		**2**		**3**		**4**		**5**		**6**		**7**		**8**	
J_k																	
$k < J_k$		***Y***		***Y***		***Y***		***Y***		***Y***		***Y***		***Y***		***Y***	
		N		***N***		***N***		***N***		***N***		***N***		***N***		***N***	

1 (iii) $A \cup B$

$$= \left\{ \quad \right\} \cup \left\{ \quad \right\} = \left\{ \quad \right\}$$

1 (iv) $A \cap B$

$$= \left\{ \quad \right\} \cap \left\{ \quad \right\} = \left\{ \quad \right\}$$

No points on this page if answers to (iii) and (iv) are incorrect as they act as checks for (i) and (ii).

6 (1b) Compute the following, **showing all work including trees on the on pages that follow, properly numbered,** and record **ONLY** the answers on this page.

Your J-number $JN :=$ _____ $:=$ $(J_k | k \in 1..8)$

$$\mathcal{U} := \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

0 (P) $P := \{J_k \in 0..9 \mid Evn(J_k)\} := \{\quad\}$

k		**1**		**2**		**3**		**4**		**5**		**6**		**7**		**8**	
J_k																	
$Evn(J_k)$		***Y***		***Y***		***Y***		***Y***		***Y***		***Y***		***Y***		***Y***	
		N		***N***		***N***		***N***		***N***		***N***		***N***		***N***	

0 (P^c) $P^c = \mathcal{U} \backslash P = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\} \backslash \{\quad\} = \{\quad\}$

0 (Q) $Q := \{k \in 1..8 \mid Evn(J_k)\} := \{\quad\}$

k		**1**		**2**		**3**		**4**		**5**		**6**		**7**		**8**	
J_k																	
$Evn(J_k)$		***Y***		***Y***		***Y***		***Y***		***Y***		***Y***		***Y***		***Y***	
		N		***N***		***N***		***N***		***N***		***N***		***N***		***N***	

0 (Q^c) $Q^c = \mathcal{U} \backslash Q = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\} \backslash \{\quad\} = \{\quad\}$

Two of these problems also occur on the next page where they carry points. Please check that you do not make a mistake in copying them.

1 (i) $P^c = \mathcal{U}\backslash P = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\} \backslash \{\quad\} = \{\quad\}$

1 (ii) $Q^c = \mathcal{U}\backslash Q = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\} \backslash \{\quad\} = \{\quad\}$

1 (iii) $P \cap (Q^c)$ **(same answer as (iv), otherwise no points on (ii), (iii) & (iv))**

$= \{\quad\} \cap \{\quad\} = \{\quad\}$

1 (iv) $P\backslash Q = \{\quad\} \backslash \{\quad\} = \{\quad\}$

1 (v) $(P^c) \cap Q$ **(same answer as (vi), otherwise no points (i), (iv) and (v))**

$= \{\quad\} \cap \{\quad\} = \{\quad\}$

1 (vi) $Q\backslash P = \{\quad\} \backslash \{\quad\} = \{\quad\}$

1(vii) $\left(P\backslash Q\right) \cap \left(Q\backslash P\right) = \{\quad\} \cap \{\quad\} = \{\quad\}$

No points on this page if the answer to (vii) is incorrect. The problems are interrelated and serve as checks on whether the work is being done with sufficient care and attention. Such built-in checks will occur on all exams. I am declaring this feature here but will not do so on other exams.

5 (1c) Circle correct choices from among Y, N, Proof, and Witness and provide below a proof or witness as the case may be:

1 (i) $\forall x \in \mathbb{N} \quad \exists y \in \mathbb{Z} \quad \left(x - y = x\right)$

$Y \quad N \quad Pf \quad W$

Why?

1 (ii) $\exists x \in \mathbb{N} \quad \forall y \in \mathbb{Z} \quad \left(x - y = -y\right)$

$Y \quad N \quad Pf \quad W$

Why?

1 (iii) $\forall x \in \mathbb{N} \quad \exists y \in \mathbb{Z} \quad \left(x + y = 0\right)$

$Y \quad N \quad Pf \quad W$

Why?

1 (iv) $\exists x \in \mathbb{Z} \quad \forall y \in \mathbb{N} \quad \left(xy \leq 0\right)$

$Y \quad N \quad Pf \quad W$

Why?

1 (v) $\exists x \in \mathbb{Z} \quad \exists y \in \mathbb{Z} \quad \left(x + y + xy = 1\right)$

$Y \quad N \quad Pf \quad W$

Why?

6 (2a) Answer the following:

1 (i) $1 \in \left\{0, \{1\}\right\}$ Y N Pf W

Why?

1 (ii) $\{1\} \in \left\{0, \{1\}\right\}$ Y N Pf W

Why?

1 (iii) $1 \subseteq \left\{0, \{1\}\right\}$ Y N Pf W

Why?

1 (iv) $\{0\} \subseteq \left\{0, \{1\}\right\}$ Y N Pf W

Why?

1 (v) $\left\{\{0\}\right\} \in \left\{0, \{1\}\right\}$ Y N Pf W

Why?

1 (vi) $\left\{\{0\}\right\} \subseteq \left\{0, \{1\}\right\}$ Y N Pf W

Why?

6 (2b) **Show work pages that follow, properly numberd, and record only the answers on this page**. Define the sequence of sets: $(S_n | n \in \mathbb{N})$ recursively as follows:

Base case: (BC_0) $S_0 \coloneqq \{\}$

Recursive step:(RcS) $\forall n \in \mathbb{N}$ $S_{n+1} \coloneqq \left(\mathcal{P}\left(S_n\right)\right) \setminus \left\{S_n\right\}$

1 (i) Compute: S_1 =

1 (ii) Compute: $\nu(S_1)$ =

1 (iii) Compute: S_2 =

1 (iv) Compute: $\nu(S_2)$ =

1 (v) Formulate a conjecture for the value of $\nu(S_n)$in terms of n.

$\nu(S_n)$ =

1 (vi) Verify your conjecture on the facing side for: $n = 3$.

5 (2c) Circle correct choices from among Y, N, Proof, and Witness and provide a proof or witness as the case may be, with complete reasoning, as in the notes on the facing side:

2 (i) $\forall X, Y, Z \in \mathcal{U} \qquad \left(X \backslash Y\right) \times Z = \left(X \times Z\right) \backslash \left(Y \times Z\right)$

$Y \qquad N \qquad Pf \qquad W$

3 (ii) $\forall X, Y, Z \in \mathcal{U} \qquad Z \times \left(X \backslash Y\right) = \left(Z \times X\right) \backslash \left(Z \times Y\right)$

$Y \qquad N \qquad Pf \qquad W$

6 (3a) Compute the following, **showing all work on pages that follow, properly numbered,** and record **ONLY** the answers on this page.
$J_k \in 0..9$

Your J-number $JN :=$ _____ $:=$ $(J_k | k \in 1..8)$

2 (i) $E := \left\{ k \in 1..8 \middle| Prm(k + J_k) \right\} = \left\{ \quad \right\}$

k		**1**		**2**		**3**		**4**		**5**		**6**		**7**		**8**	
J_k																	
$k + J_k$																	
$Prm(k + J_k)$		***Y***		***Y***		***Y***		***Y***		***Y***		***Y***		***Y***		***Y***	
		N		***N***		***N***		***N***		***N***		***N***		***N***		***N***	

2 (ii) $F := \left\{ k \in 1..8 \middle| Odd(k + J_k) \right\} = \left\{ \quad \right\}$

k		**1**		**2**		**3**		**4**		**5**		**6**		**7**		**8**	
J_k																	
$k + J_k$																	
$Odd(k + J_k)$		***Y***		***Y***		***Y***		***Y***		***Y***		***Y***		***Y***		***Y***	
		N		***N***		***N***		***N***		***N***		***N***		***N***		***N***	

1 (iii) $E \cap F = \left\{ \quad \right\}$

1 (iv) $\mathcal{P}\left(E \cap F \right) = \left\{ \quad \right\}$

6 (3b) Circle correct choices from among Y, N, Proof, and Witness and provide below a proof or witness as the case may be:

3 (i) $\exists X, Y \in \mathcal{U}$ $\left(X \backslash Y = Y \backslash X\right)$

Y N Pf W

3 (ii) $\forall X, Y \in \mathcal{U}$ $\left(X \backslash Y = X \backslash (X \cap Y)\right)$

Y N Pf W

1. 5 (3c) Define $\forall X, Y \in \mathcal{U} \quad X\&Y \coloneqq \left((X^c)\backslash(Y^c)\right) \cup \left((Y^c)\backslash(X^c)\right)$

Circle correct choices from among Y, N, Proof, and Witness and provide below a proof or witness as the case may be:

2 (i) $\forall X \in \mathcal{U} \quad \forall Y \in \mathcal{U} \quad \left(X\&Y = Y\&X\right)$

$Y \quad N \quad Pf \quad W$

3 (ii) $\forall X \in \mathcal{U} \quad \forall Y \in \mathcal{U} \quad \forall Z \in \mathcal{U} \quad \left((X\&Y)\&Z = X\&(Y\&Z)\right)$

$Y \quad N \quad Pf \quad W$